

Historical Drilling Incident Review: Montara Wellhead Platform (2009)

1. Operational Context

On August 21, 2009, an uncontrolled release of oil and gas occurred at the Montara wellhead platform in the Timor Sea, offshore Western Australia, during batch drilling operations on well H1-ST1.

2. Primary Containment Failure

The 9-5/8 inch casing shoe cement job had failed during installation. A primary barrier was not successfully established, allowing reservoir fluids to migrate up the casing annulus. No pressure testing was performed on the secondary mechanical barriers prior to suspending operations.

3. Pressure Transient & Kick Dynamics

During the removal of the corrosion cap, an uncontrolled influx of light crude and reservoir gas occurred. Wellbore hydrostatic pressure was insufficient due to an under-balanced fluid column of light brine without proper weighting material.

4. Regulatory & Remediation Findings

The Montara Commission of Inquiry noted systemic deficiencies in barrier policy enforcement, casing centralization, and well suspension verification. The blowout was finally killed via relief well drilling after five attempts.